Symmetric point sets and the Erdős–Szekeres problem for heptagons

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Abstract

Let $ES(7)$ be the least n such that every n points in general position in the plane contain 7 points in convex position; the Erdős–Szekeres conjecture predicts $ES(7) = 33$. We show, with computer-assisted proofs, that every 33-point set in general position with a threefold rotational symmetry contains a convex heptagon, and that every mirror-symmetric set of at least 33 points does. Together with explicit examples, the largest threefold-symmetric and mirror-symmetric sets without a convex heptagon have exactly 31 and 32 points. Combined with the fourfold result of Subercaseaux, Mackey, Qian and Heule and a fivefold exclusion, the only symmetric sets of at least 33 points that could still avoid a convex heptagon are centrally symmetric sets of even size at least 34. As companion material we record exact maxima for other symmetry types, with their antecedents, and an explicit centrally symmetric 32-point example, which is a special case of a construction of Valtr described by Baek and Balko. None of this settles $ES(7)$ or improves its upper bound.

1 Introduction

Erdős and Szekeres [3] proved that $ES(k)$ is finite and constructed 2^{k-2} points in general position with no k in convex position [4]; they conjectured $ES(k) = 2^{k-2} + 1$. The conjecture holds for $k \le 6$, the case $k = 6$ by a computer search of Szekeres and Peters [5], and $ES(k) \le 2^{k+o(k)}$ by Suk [6]. The first open case is $k = 7$.

Symmetric configurations are a natural restricted search space. Subercaseaux, Mackey, Qian and Heule [7] built rotational symmetry into SAT encodings, found symmetric extremal sets for $k = 6$, showed that no real fourfold symmetric 32-point set avoids a convex heptagon, and asked for a realized twofold symmetric 32-point example. Here we exclude symmetric counterexamples to $ES(7) = 33$ except possibly centrally symmetric ones of even size at least 34.

Throughout, *general position* means no three points on a line, and a *k-gon* is a set of k points in convex position. Computations are described with their trust base in Section 6; directory names refer to `cloud/es7-2026-10-19/results` on the repository branch `es7-cloud`.

2 Which symmetries are possible

The symmetry group of a finite set fixes its centroid, so it is a cyclic group C_s of rotations or a dihedral group. For a rotation of order s about c, every orbit other than $\{c\}$ has exactly s points and is the vertex set of a regular s-gon.

Lemma 1. *Let S be in general position and invariant under a rotation ρ of order $s \ge 2$ about c. (a) If $s \ge 7$ and $S \ne \{c\}$, then S contains a 7-gon. (b) If s is even and $|S| \ge 3$, then $c \notin S$ and s divides $|S|$. (c) If s is odd, then $|S| \equiv 0$ or $1 \pmod{s}$.*

Proof. (a) An orbit is a convex s-gon. (b) $\rho^{s/2}$ is the half-turn, so $c, p, \rho^{s/2}(p)$ are collinear for every p; hence $c \notin S$ and all orbits have size s. (c) At most the centre is a fixed point. □

3 Threefold symmetry

Theorem 2 (computer-assisted)**.** *Every set of* 33 *points in general position that is invariant under a rotation of order* 3 *contains* 7 *points in convex position. The largest threefold-symmetric set in general position without a convex heptagon has exactly* 31 *points.*

A threefold set has $3m$ or $3m+1$ points, so one with at least 33 points contains 11 whole orbits of size 3, a 33-point threefold set without its centre; 32 is not a possible size. An exact 31-point example (centre included) is `SYMMAX/c3_31points.txt`.

For the 33-point case, a projective normal form of the labelling (`C3-S/soundness.md`) gives a CNF with 168,399 variables and 4,154,921 clauses. Its unsatisfiability was established by splitting on the convex layers and on orientation bits of symmetric residues into 116,883 certified pieces. A reflection of the $H6$ strata (Lemma H6M) leaves 910 depth-2 roots. For each piece, CaDiCaL produced an LRAT proof that was accepted at solve time both by `lrat-check` and by the formally verified checker `cake_lpr` [8]; these proofs were then deleted. Two independently written composers recompute the CNF hash of every certified piece from the base formula and find all 910 roots covered (`C3FINAL`); an independent audit replayed this composition. In addition, 2,120 randomly drawn pieces were re-solved from scratch with fresh proofs and both checkers (`C3SPOT`).

4 Mirror symmetry

Theorem 3 (computer-assisted)**.** *Every mirror-symmetric set of at least* 33 *points in general position contains* 7 *points in convex position. The largest mirror-symmetric set in general position without a convex heptagon has exactly* 32 *points.*

Assembly. By general position at most two points lie on the axis. If none does, the size is even, hence at least 34, and 17 mirror pairs form an axis-free 34-point mirror set: this is excluded by Theorem 5. If one or two do, 16 mirror pairs and one axis point form a 33-point mirror set with exactly one point on the axis: this is excluded by Theorem 4. An exact 32-point example with 16 pairs is `SYMMAX/mirror_32points.txt`. □

Theorem 4 (computer-assisted)**.** *A mirror-symmetric set of* 33 *points in general position with exactly one point on the axis contains a* 7*-gon.*

Theorem 5 (computer-assisted)**.** *A mirror-symmetric set of* 34 *points in general position with no point on the axis contains a* 7*-gon.*

Theorem 4 has two independent proofs. The first (`WRITEUP/MIRROR.md`) uses a signotope encoding whose soundness, normal forms, case split and leaf cover are verified in Lean 4; its roughly 75,000 CNF facts carry recorded `cake_lpr` verdicts, and their proof files were not retained. The second, and the proof of Theorem 5, use the encoding below; each is a single CNF with a retained LRAT proof accepted by `lrat-check` and `cake_lpr` (`SYMZ`).

The encoding. Put the axis on the y-axis, write the pairs as $L_i = (-a_i, b_i)$, $R_i = (a_i, b_i)$ with $a_i > 0$, use homogeneous vectors $v_p = (1, x_p, y_p)$, and add one element Z with vector $(0, 1, 0)$, the horizontal point at infinity. The chirotope $\chi = \operatorname{sign}\det$ of the set and Z satisfies: (i) $\det(v_Z, v_p, v_q) = y_p - y_q$, so the zero triples are exactly $\{Z, L_i, R_i\}$; (ii) the reflection acts by $\operatorname{diag}(1, -1, 1)$, which has determinant -1 and negates v_Z, so it reverses the orientation of triples of set points and preserves that of triples containing Z; (iii) $v_{R_i} - v_{L_i} = 2a_i v_Z$ with $a_i > 0$, so $\chi(Z, L_i, q) = \chi(R_i, L_i, q) = \chi(Z, R_i, q)$; (iv) the three-term Grassmann–Plücker relations hold, and the vectors are acyclic, since $w + \varepsilon x$ is positive on all of them for small $\varepsilon > 0$; (v) numbering the pairs by decreasing height gives $\det(v_Z, v_{L_i}, v_{L_j}) = b_i - b_j > 0$ for $i < j$. The CNF encodes (i)–(v), with variables for the classes of identified triples, and the absence of a 7-gon through the four-point criterion. Every real configuration therefore gives a satisfying assignment, so unsatisfiability excludes all real configurations. The converse fails: a satisfying assignment need not be realizable, and Z imposes no metric condition.

For 34 points the CNF has 838,032 variables and 16,642,456 clauses and is refuted by CaDiCaL in 443 s; for 33 points with one axis point it has 695,928 variables and 14,080,656 clauses and is refuted in 964 s. The encoding was tested on real symmetric sets and on 3,300 random symmetric sets (satisfiable exactly when no K-gon exists, no mismatch), and it reproduces the mirror hexagon bound of Proposition 13.

Corollary 6. *Assume Theorems 2 and 3, Theorem 8 below and the fourfold exclusion of [7, §6.2]. Then a set of at least* 33 *points in general position without a convex heptagon either has no nontrivial symmetry or has symmetry group exactly the half-turn group, and then it has an even number of points, at least* 34*.*

Proof. Reflections are excluded by Theorem 3. Rotations of order $s \geq 7$ are excluded by Lemma 1(a), and orders 3, 4 and 5 by deleting whole orbits; order 6 contains order 3. The half-turn remains, and by Lemma 1(b) its centre is not a point of the set. □

5 Companion results

Lemma 7. *Let S be in general position and invariant under a rotation ρ of order $m \geq 4$ about a point $c \notin S$, and suppose S has no K points in convex position for some $K \leq 2m$. Then every convex layer of S is a single orbit.*

Proof. The vertex set V of $\operatorname{conv}(S)$ is invariant under ρ, so it is a union of orbits of m points each, and it is in convex position; hence $|V| < K \leq 2m$ and V is one orbit. Apply the same argument to $S \setminus V$. □

Theorem 8 (computer-assisted)**.** *Every set of at least* 35 *points in general position with a rotation of order* 5 *contains* 7 *points in convex position.*

Such a set contains a 35-point fivefold set without centre, which by Lemma 7 would be seven nested regular pentagons. The CNF of `tools/symmetric/symcnf.py` for this case (an auxiliary centre, in general position after rotating each orbit slightly) is refuted in 415 s, and a second normal form using Lemma 7 in 54 s; both LRAT proofs are retained and accepted by both checkers (`SYM5`).

Remark 9. Theorem 8 also follows from an earlier reported computation. Bogdan [2] reports an unsatisfiable run for 33 points with six convex pentagonal layers and three unconstrained interior points, and deleting two points of the innermost pentagon of a 35-point fivefold set gives exactly that configuration. No proof files for that run are public, so we treat it as a prior reported claim rather than an independently verified theorem.

Proposition 10. *Let $N(\Gamma)$ be the largest size of a set in general position with symmetry Γ and no 7 points in convex position. Then $N(C_3) = 31$ and $N(\mathit{mirror}) = 32$ (Theorems 2 and 3), $N(C_6) = 30$, $N(C_4) = 28$, $N(C_5) = 31$ and $N(C_2) \geq 32$, with $N(C_2) = 32$ if $ES(7) = 33$.*

Proof. Lower bounds are exact examples in `SYMMAX` and `SYM2`, found by simulated annealing with exact arithmetic and checked by a dynamic program and by brute force over all 7-subsets. A sixfold set of 36 points contains a threefold 33-point set, so $N(C_6) = 30$ follows from Theorem 2. The fourfold bound is due to [7, §6.2], who also report realizable 28-point restrictions. The fivefold bound is Theorem 8 and Lemma 1(c). □

An explicit centrally symmetric example. Baek and Balko [1, Definition 13, Lemma 14 and §6.1] describe a construction of Valtr that blows up an arbitrary set of $k-1$ points in general position into 2^{k-2} points without a convex k-gon, using clusters with cup and cap parameters $x_i = i-1$ and $y_i = k-i-1$. For odd k, a centrally symmetric base with clusters paired by the half-turn, which exchanges cups and caps and the two parameters, gives a centrally symmetric set. The following is an explicit case of this, rotated by a quarter-turn, with a direct proof. Sort points by x-coordinate. An *a-cup* is a sequence of a points whose consecutive slopes increase, an *a-cap* one whose slopes decrease; let $G(a,b)$ be the classical set of $\binom{a+b-4}{a-2}$ points in general position with no a-cup and no b-cap [3]. The half-turn $h(p) = -p$ exchanges cups and caps.

Proposition 11. *Let $k \geq 3$ be odd, $n = k-2$ and $x_i = 2i - n$. For $i < n/2$ place a copy of $G(i+2, k-i)$, flattened by $(x,y) \mapsto (\varepsilon x, \varepsilon^2 y)$, at (x_i, x_i^3), and for $i > n/2$ place the image under h of the copy at index $n-i$. For small ε the union S is a centrally symmetric set of 2^{k-2} points in general position without k points in convex position.*

Proof. Cluster B_i has no $(i+2)$-cup and no $(k-i)$-cap, and $|S| = \sum_i \binom{n}{i} = 2^{k-2}$. Three points of the cubic at $x = a, b, c$ are collinear iff $a+b+c = 0$, impossible for odd numbers. The slope between clusters $a < b$ is close to $\sigma(a,b) = x_a^2 + x_a x_b + x_b^2 \geq 1$, slopes inside clusters tend to 0, and $\sigma(b,c) - \sigma(a,b) = (x_c - x_a)(x_a + x_b + x_c)$; so for small ε the set is in general position and consecutive segments between clusters turn according to the sign of $x_a + x_b + x_c$.

Let $P \subseteq S$ be in convex position with leftmost point in B_l and rightmost point in B_r. If $l = r$ then $|P| \leq k-2$. Otherwise the lower chain is a cup in B_l (at most $l+1$ points), then single points in clusters $c_1 < \cdots < c_t$ strictly between l and r, and the upper chain is single points in clusters $d_1 < \cdots < d_s$ strictly between, then a cap in B_r (at most $k-r-1$ points). The turn conditions reduce to $x_l + x_{c_1} + x_{c_2} > 0$ (with $c_2 = r$ if $t = 1$) and $x_{d_{s-1}} + x_{d_s} + x_r < 0$ (with $d_{s-1} = l$ if $s = 1$). If $t, s \geq 1$ and $c_1 \leq d_s$, comparing the two sums termwise gives a contradiction; so the clusters used by the two chains are distinct, $t + s \leq r - l - 1$, and $|P| \leq (l+1) + (k-r-1) + t + s \leq k-1$. □

For $k = 7$ this gives 32 points; an integer instance with coordinates of absolute value at most 1,250,000 is in `SYM2`, checked by a dynamic program and by brute force over all $\binom{32}{7}$ subsets. It answers the concrete realization task of [7]; it is not a new general construction method.

Hexagons. For $k = 6$ the largest symmetric sets without a convex hexagon have 16 points for twofold, fourfold and fivefold rotations (the fourfold and fivefold examples are in [7]; a fourfold set is twofold), 13 points for threefold rotations (15 and 16 points are excluded by retained LRAT proofs; 16 agrees with [7]), and 12 points for a reflection, by the following elementary bound.

Lemma 12. *Let S be in general position and symmetric under the reflection μ in a line ℓ, and let $p_1, \ldots, p_j$ be points of S strictly on one side of ℓ, ordered by their projections to ℓ, forming*

a chain in which every middle point is farther from ℓ than the chord of its neighbours. Then $\{p_1, \dots, p_j, \mu p_1, \dots, \mu p_j\}$ is in convex position.

Proof. Traverse $p_1, \dots, p_j, \mu p_j, \dots, \mu p_1$. Along each chain the turns have a common sign because the chain bulges away from ℓ. At p_j the next edge is perpendicular to ℓ, and the turn has the same sign, its cross product being $2\,\mathrm{dist}(p_j, \ell)$ times the projection gap. The same holds at p_1, and the mirror images give the rest. □

Proposition 13. *Let $k = 2j$ be even, and let S be a mirror-symmetric set in general position with no k points in convex position. Then $|S| \le 2\binom{3j-4}{2j-2} + 2$; in particular $|S| \le 12$ for $k = 6$ and $|S| \le 58 < 2^6$ for $k = 8$.*

Proof. With the projection to ℓ as abscissa and the distance to ℓ as height, Lemma 12 says the points on one side have no j-cap; they have no k-cup, since a cup is in convex position. By the cups-caps theorem [3] there are at most $\binom{3j-4}{2j-2}$ of them, and at most two points lie on ℓ. □

6 Trust base

Geometric soundness of every encoding rests on written proofs, which were audited independently: the threefold normal form and encodings, the fivefold encodings and Lemma 7, and the mirror encoding of Section 4. *Lean-verified* are only the reductions of the first proof of Theorem 4. *External certificate checks* use `lrat-check` (with a patch for a known upstream defect) and `cake_lpr`, a checker verified in CakeML; the checkers themselves are trusted. For the threefold theorem the per-piece proofs are *recorded verdicts* only: they were checked at solve time and then deleted; the composition and 2,120 random pieces were *replayed*, which is evidence but not an exhaustive recheck. The mirror and fivefold proofs are retained in a compressed archive with their hashes, and the 34-point mirror proof was replayed by an independent audit.

7 What remains open

$ES(7)$ itself is open, and no unrestricted bound is improved here. By Corollary 6, the remaining symmetric route to a counterexample is a centrally symmetric set of even size at least 34; excluding 34 points would exclude all of them. For the analogous encoding with the centre c as an element ($v_c = (v_p + v_{-p})/2$, so $\chi(c, p, q) = \chi(-p, p, q)$), a split by the convex hull into 136 cases has a certified cover. All 120 hexagonal-hull cases are refuted, each with a retained proof, so a centrally symmetric 34-point counterexample would have a parallelogram hull. The parallelogram cases are open; nested parallelograms, the layer structure of the extremal 32-point sets, are the hard core (`C2C34`). An abstract centre imposes no metric midpoint condition and no realizability: the fourfold 32-point formula of the same kind is satisfiable, as [7] also observed for its own model, although no real set exists. Only an unsatisfiable answer, with a justified case cover, would settle the case.

References

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